

Structural Predictors of Chronic Absenteeism in NYC Schools

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Problem

Chronic absenteeism remains a structural barrier across NYC's ~900k-student public school system, persisting beyond the Pandemic

Socioeconomic and demographic drivers continue to shape student outcomes at scale and warrant a school-level analytical framework

KEY OBSERVATIONS

- I. **Severity of Student Impact**
Chronic absenteeism is associated with lower academic achievement and reduced graduation likelihood, with effects that persist into adulthood and shape long-term educational trajectories

- II. **Policy-level Invisibility**
Without school-level analysis, structural causes of chronic absenteeism remain invisible to district-level policy and resource allocation decisions, even as they shape attendance outcomes at scale.

- III. **Empirical Consensus on Measurement**
Risk Thresholds cluster in the 3-7% range, supporting treatment of chronic absenteeism as a continuous rate rather than a binary classification

Background Research

PRIOR WORK

Mukli & Rista (2023)

Studied 500 university students at one institution - limited to personal-life predictors and not generalizable

Vald (NY Trends 2018-19)

Analyzed daily attendance dips against weather and holidays - pre-pandemic with no structural drivers

Wu, Weiland, and Staines (2026)

Defined the threshold itself, but did not model it across a real district



GAP

Existing work doesn't address...

- why chronic absenteeism exists institutionally
- structural drivers in a city-wide system

We aim to bridge the gap by testing school-levels NYC prediction with temporal validation and interpretable structural features.

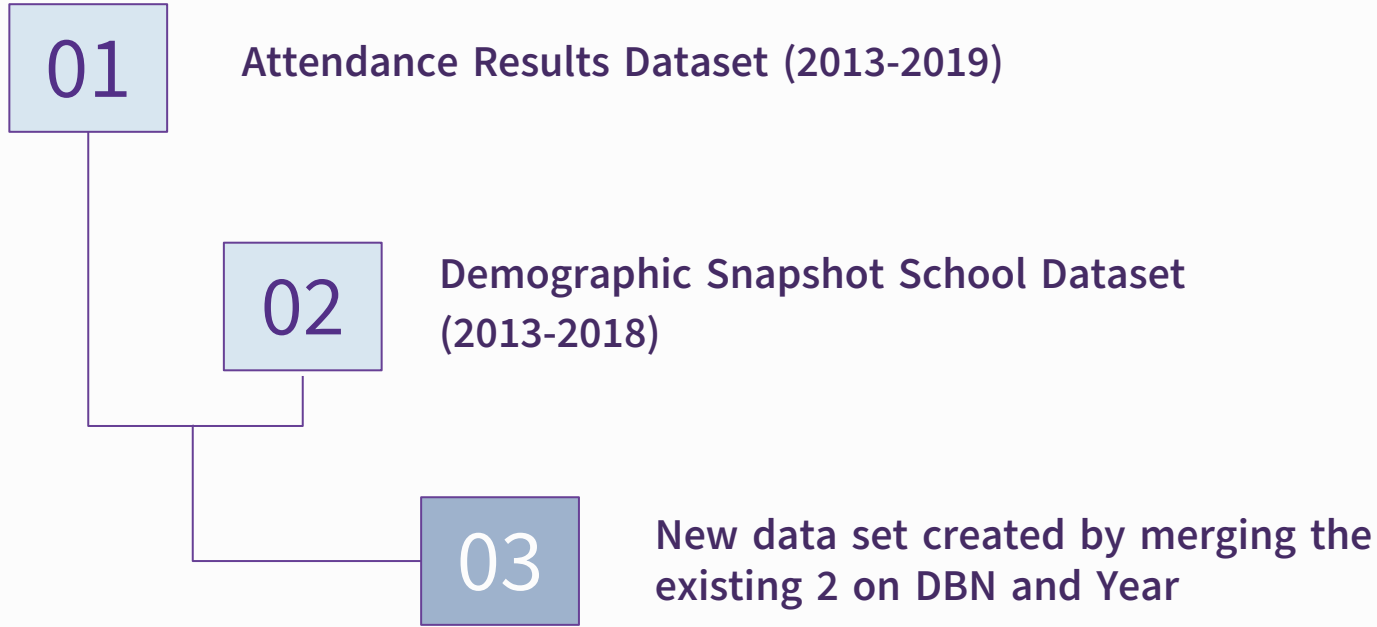
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Predict the percentage of chronically absent students at each NYC public school, given its structural profile, modeled as a continuous rate.

OUR GOAL

Our Dataset

We draw from two publicly available NYC DOE Open Data sources.



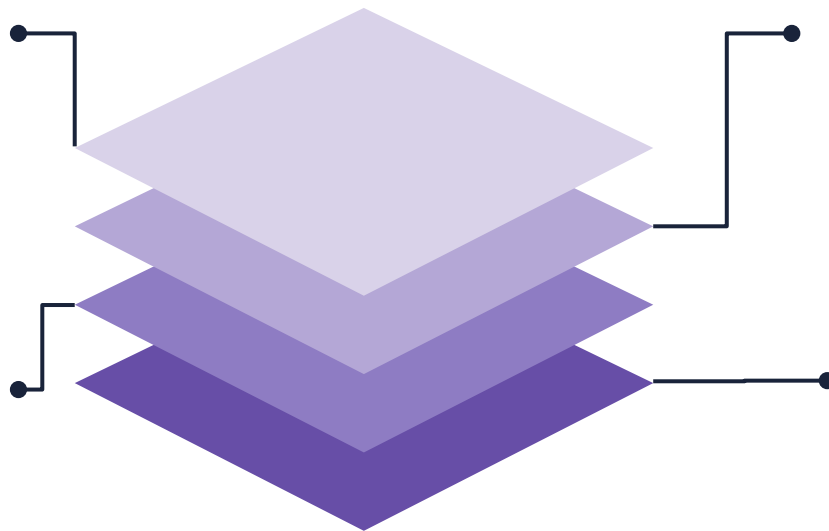
Dataset Modification

2013-2018

Years Spanned

7,607

Total Observations



50

Features

3

Splits

Key Features

FOUR STRUCTURAL PREDICTORS

ENI**

*Economic Need
Index*

Poverty

ELL

*English Language
Learners*

SWD

*Students with
Disabilities*

DATA CONSTRAINTS

1493

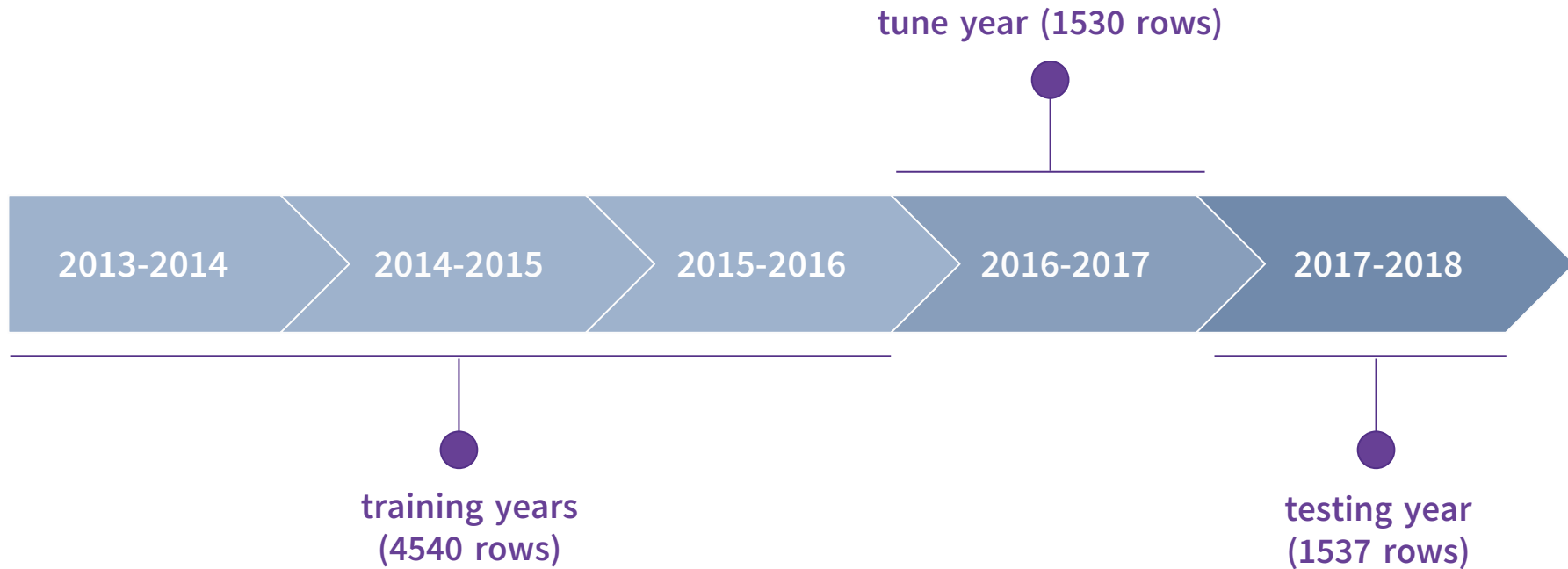
Missing ENI values

in the 2013-14 academic year only

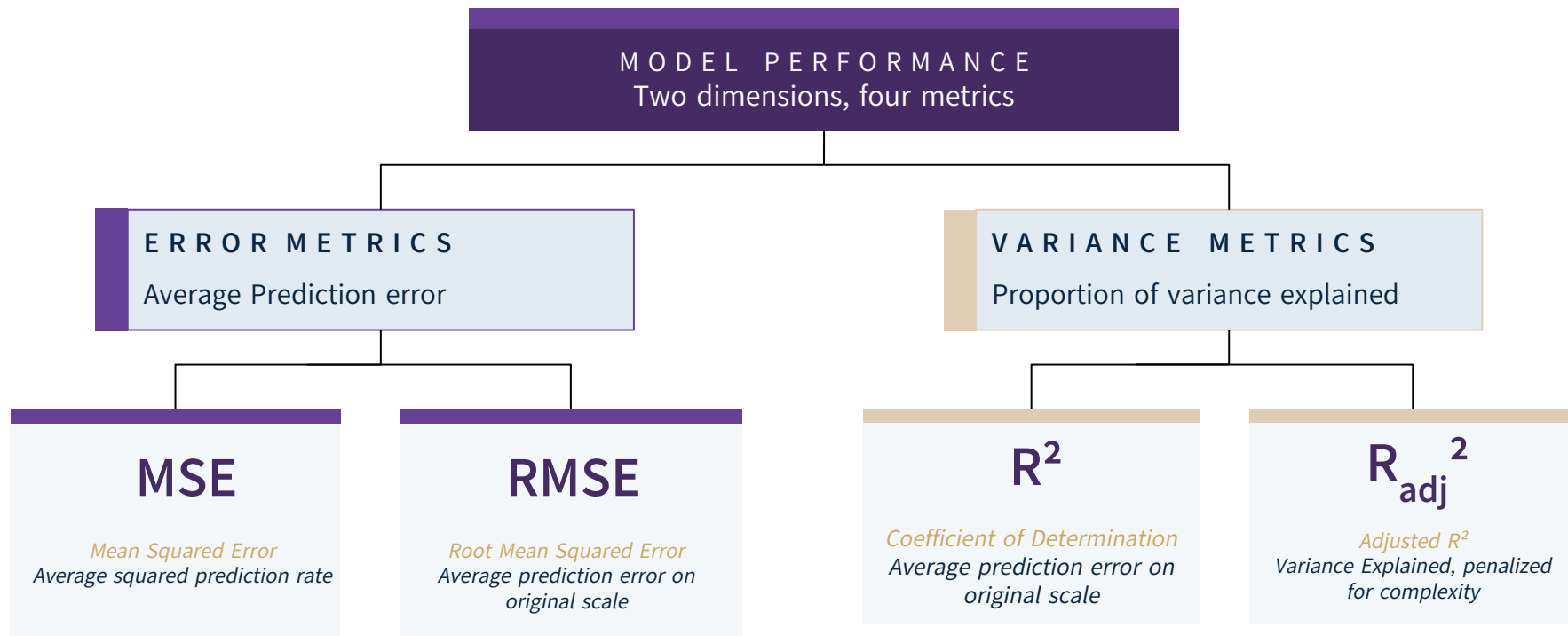
Why we proceeded: We believed the missing ENI values to be mitigable due to our train split containing ENI for 2014-2015 and 2015-2016 on top of having 3 other predictors.

Data Splitting

Temporal (60/20/20) Split



Evaluation Metrics



*MSE and RSME capture average prediction error with
RSME being more interpretable*

*R^2 and Adjusted R^2 measure explained variance, with
the latter penalizing model complexity*

Baseline Methods

Multiple Linear Regression

An OLS regression of % 'Chronically Absent' on four structural predictors.

- Economic Need Index (ENI)
- % Poverty
- % English Language Learners (ELL)
- % Students with Disabilities (SWD)

Polynomial Regression

- Used to determine the best predicting model order
- Extension of MLR baseline by replacing the linear ENI term with a polynomial degree ($j = 1...10$)
- ENI has by far the highest t-value (34.381), suggesting statistical significance

ML Methods

Random Forest

Implemented RF on four structural predictors using 5-fold cross-validation

- Variables Sampled: range 1-4
- Min. Node Size: range 2-20
- Fixed trees: 500
- Goal: selecting the combination that minimizes RMSE

XGBoost

Implemented on the same four structural predictors using 5-fold cross-validation to tune three hyperparameters.

- Number of trees: 100–1000
- Tree depth: 3-8
- Learning rate: 0.001–0.1, searched on a log scale)

Diagnostics & Error Analysis

SHAP Analysis

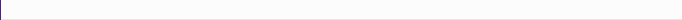
- Computed TreeSHAP values on test set for both RF and XGBoost
- Computed TreeSHAP values on test set for RF and XGBoost

Distributional Shift Validation

- Density plots; K-S test; descriptive statistics

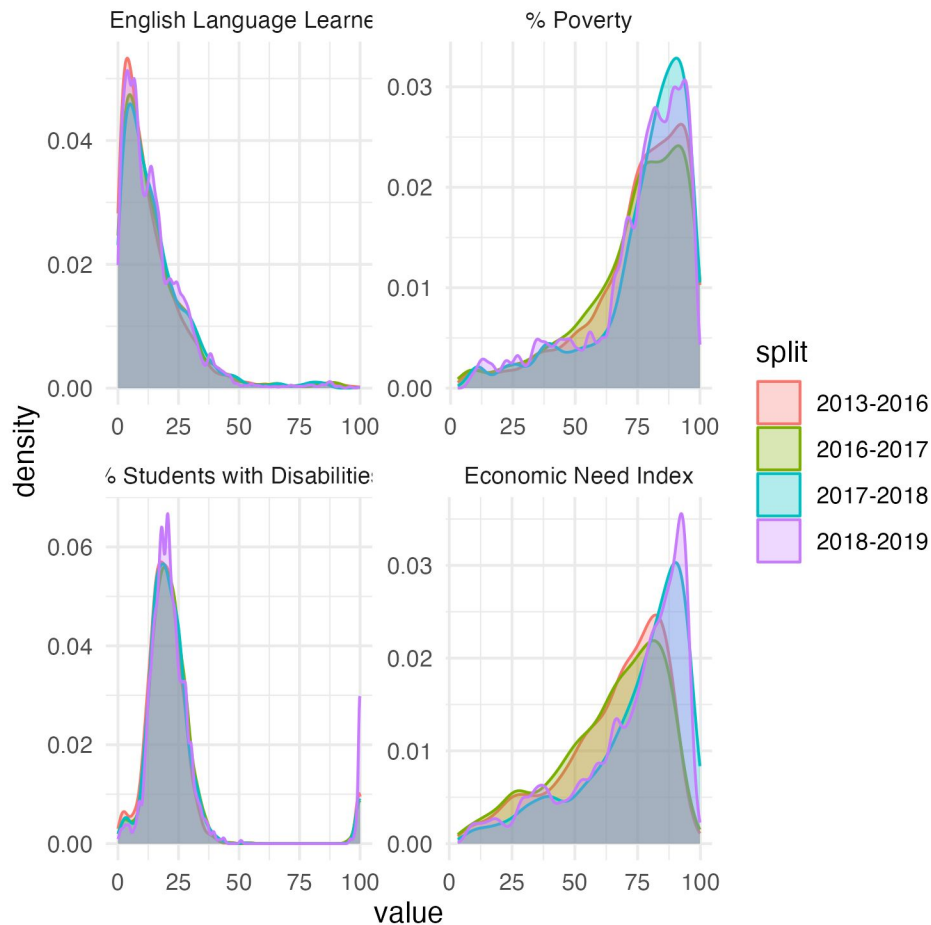
Clipping

- Clipped training ENI values at 95 to align training distribution with test distribution



Diagnostics

Experiments



Diagnostics

- Examined a random and grouped random split alongside the temporal one
 - This was done to deal with potential data leakage
 - We believed it to be a potential cause for our results but that was not the case
 - The same pattern arose in grouped random split
- Temporal Split: Tune & train error were similar but test error was much larger than both train & tune

Diagnostics Cont.

Anomaly Cause

- A new data matching process was implemented in the 2017-18 school year
- Led to an increase in both ENI and % Poverty

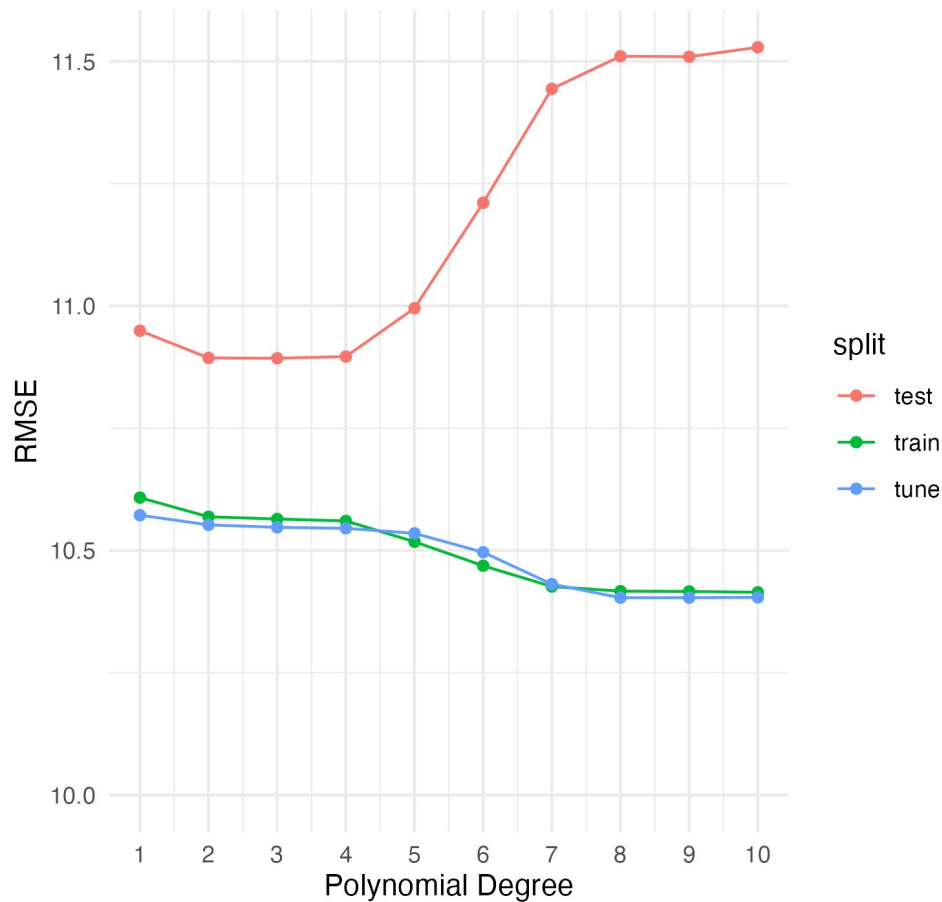


Impact on Predictors

- Distributions of ENI and % Poverty shifted after 2017-18
- ENI had a sharp mode increase toward the 95% ceiling
- K-S tests confirm ENI and % Poverty differ significantly between train/test ($D = 0.233$, $p = 0$) and tune/test ($D = 0.241$, $p = 0$)
- No substantial distributional shift for ELL and % Students with Disabilities

Multiple Linear Regression

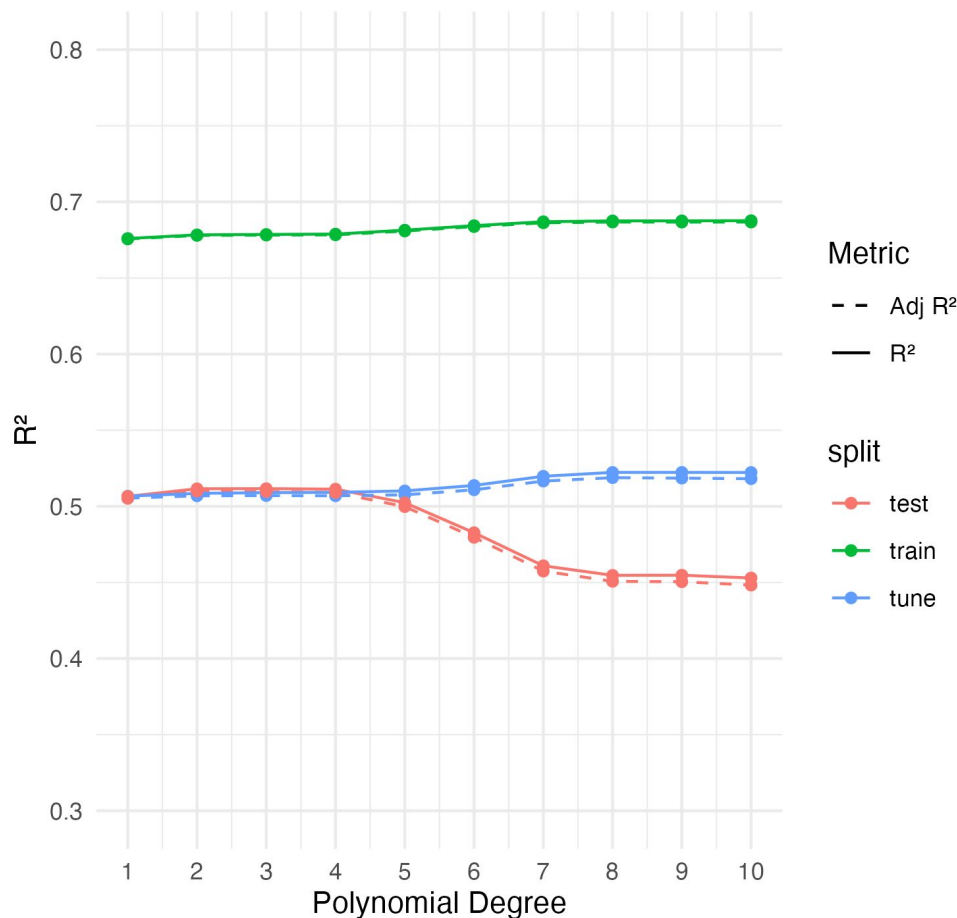
Experiments



RMSE & R^2

- MLR model yields a consistent R^2 of 0.5
- Selected predictors explain approximately half the variance in chronic absenteeism
- Train/tune RMSE remain close throughout, both decreasing with polynomial degree
- Test RMSE sits substantially higher across all degrees and rises sharply after degree 5

Train vs Test R^2 by Polynomial Degree



Train vs. Test R^2

- Tune error is minimized at degree 9
- We select degrees 3–4 as optimal
- The gap between all three splits there is smallest and model complexity remains justified before test error begins its sharp increase

Random Forest

Experiments

Table 4. RMSE values for RF.

TRAIN RMSE	TUNE RMSE	TEST RMSE
5.769	10.026	11.386

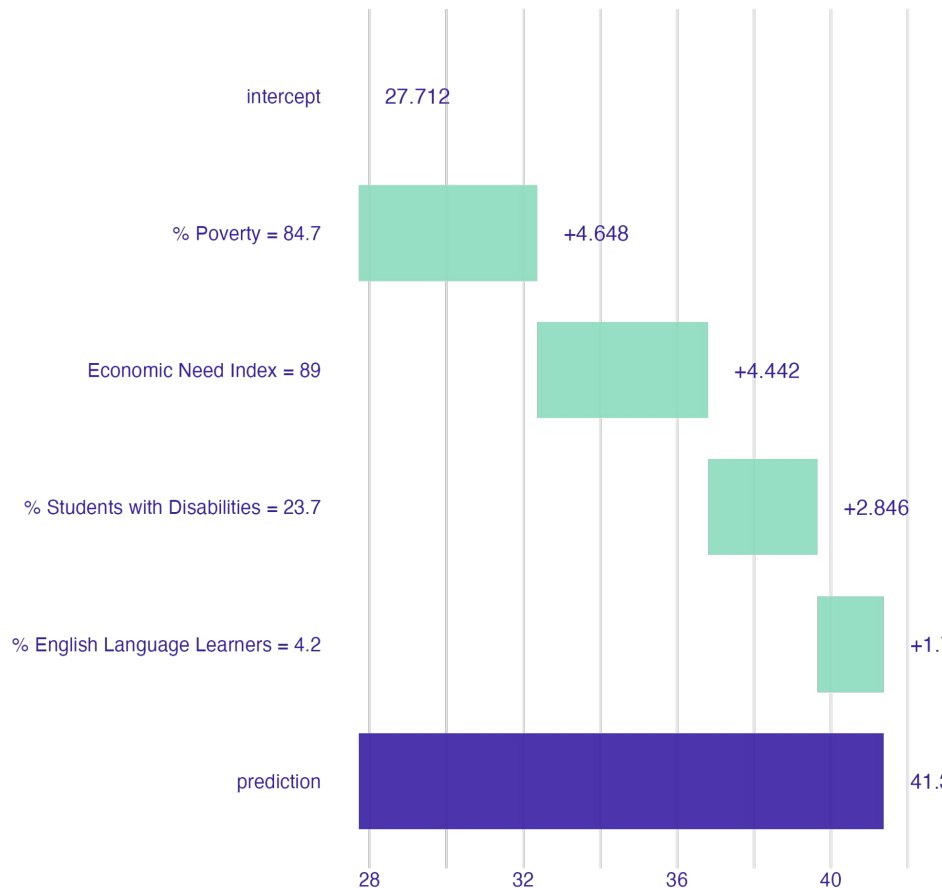
Table 5. R^2 values for RF.

	TRAIN	TUNE	TEST
R^2	0.857	0.556	0.466
R^2_{adj}	0.857	0.555	0.465

RMSE & R^2

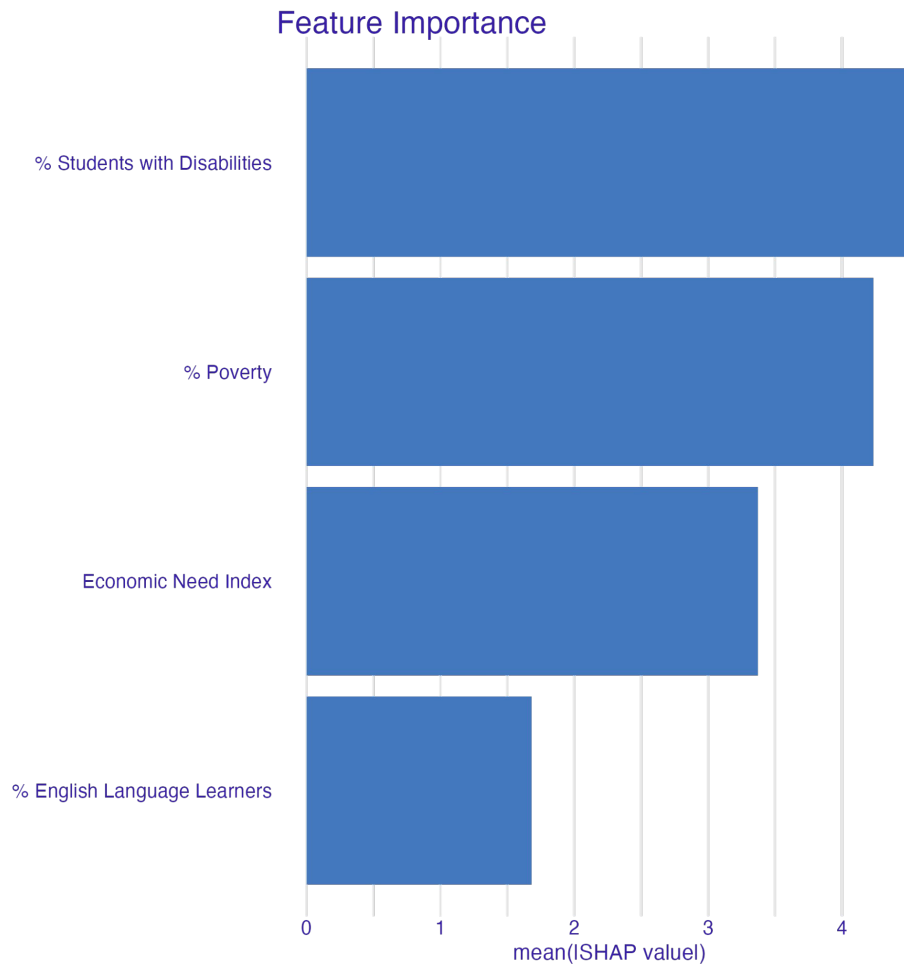
- Best hyperparameters: `mtry = 3`, `min_n = 11`
- Large minimum node size of 11 suggests that CV pushed toward greater regularization to improve generalization
- Largest train-to-test discrepancy:
 - train RMSE of 5.769 rising to 11.386 on test
 - train R^2 falling from 0.857 to 0.466
- R^2 drops to under 47% on the test set

SHAP Break-Down



SHAP Breakdown

- SHAP breakdown represents a high-need observation
 - ENI = 89
 - % Poverty = 84.7
 - % SWD = 23.7
 - % ELL = 4.2
- These data points yields a predicted chronic absenteeism rate of approximately 41%



SHAP Breakdown

- % Students with Disabilities ranks as the most important feature, with ENI falling to third
- RF's prediction that 41% of students at this school are chronically absent is substantially higher than XGBoost's prediction of 15.3%
- Possibly because of RF's higher baseline prediction of 27.7% compared to XGBoost's 2.6%,
- reflecting RF's tendency to memorize the training distribution rather than learn generalizable structure.

XGBoost

Experiments

Table 6. RMSE values for XGBoost.

TRAIN RMSE	TUNE RMSE	TEST RMSE
9.280	9.697	10.736

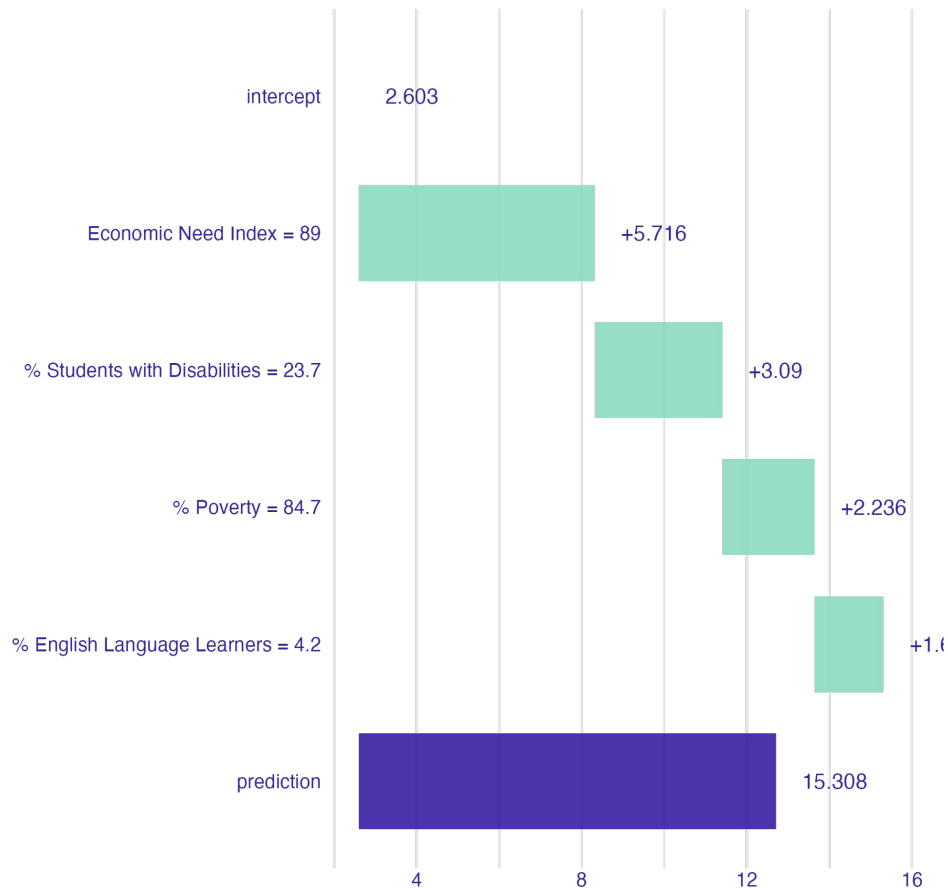
Table 7. R^2 values for XGBoost.

	TRAIN	TUNE	TEST
R^2	0.630	0.585	0.526
R^2_{adj}	0.630	0.584	0.524

RMSE & R^2

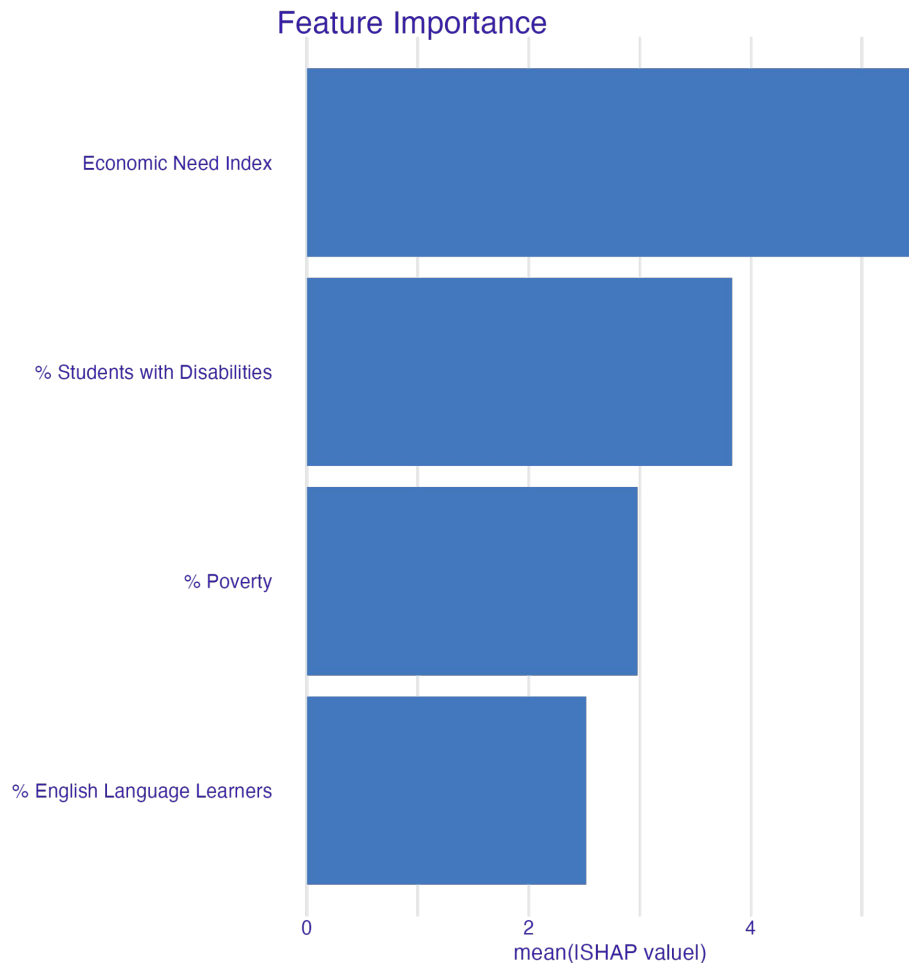
- Best hyperparameters:
 - trees = 100
 - tree_depth = 4
 - learn_rate = 0.1
- Most stable results across splits, has smallest train-to-test gap of any model tested
- Model explains between 52.6% and 63.0% of variance in chronic absenteeism across splits
- R^2 declines modestly from train to test

SHAP Break-Down



SHAP Breakdown

- same representative high-need observation as RF
 - Baseline: = 2.6%
 - ENI = +5.7
 - % Poverty = +4.6
 - % SWD = +2.8
 - % ELL = +1.6
 - Final Prediction = 15.3%
- RF and XGBoost agree on the direction and relative influence of each predictor



SHAP Breakdown

- ENI has the highest mean SHAP value
- % Students with Disabilities ranks second
- Possibly captures genuinely independent signal rather than being a redundant predictor
 - Students with disabilities may face greater barriers to attendance due to medical appointments, health-related absences, etc.
 - barriers are structurally distinct from economic need

Final Thoughts

Successes, limitations, surprises, things for the future...

OVERVIEW

Our models successfully identify the direction and magnitude of chronic absenteeism from structural variables

- XGBoost proved to be the most reliable
- ENI emerged as the dominant predictor by SHAP magnitude
 - But administrative redefinition during 2017 limited its reliability
- Amongst other predictors, % SWD proved most independently influential
 - SWD captures structural barriers to attendance distinct from economic need

FURTHERMORE

To further analyze our key limitation we ran another round of RF and XGB while capping training ENI values to 95

- XGB's RMSE was reduced to 0.52, and optimal tree depth to 3
- RF's RMSE had also improved but XGB proved more reliable



S U R P R I S E S

- Expected: RF having improved generalization due to cross-validation pushing RF toward greater regularization via larger min node size
- Reality: RF's high mtry meant it was sampling nearly the full correlated feature set at every split
 - Limiting regularization benefit
 - Leaving it more vulnerable to distributional shift

FUTURE WORK

- Future work should address...
 - Model sensitivity to changes in predictor definitions
 - Can incorporate year X ENI interaction term to explicitly model temporal instability
 - Extend framework by incorporating graduation outcomes
 - Allows the model to trace how structural absenteeism drivers translate into longer term student outcomes

Thank You

Any questions?
